OpenSourceRail

Open Engineering Platform

Published v0.6.1 software | v0.7 development reader



Wheel-supported reference and candidate-bound research. Manufacturing, physical and operating acceptance remain open.

Reader and scope

docs/BOOK.md

Reader edition: 13 November 2026. The published software release is v0.6.1. v0.7 is in development; its execution record separates completed software and numerical work from unresolved engineering and physical evidence.

OpenSourceRail develops an affordable, accessible urban railway and the open tools needed to plan, design, make, operate, maintain and renew it. Local design knowledge, manufacture and repair are central to the project. The platform connects city planning, railway engineering and manufacturing/lifecycle work; the three products must share explicit candidate and source identities.

What is available

Area	Supported scope	Boundary
Published software	v0.6.1 planning, research, Workbench and delivery tools; the clean-install Samawah journey is recorded.	Software publication does not approve hardware or train operation.
City comparison	Calculated outlines for 266 city networks and a candidate-bound Samawah station-edit comparison.	Route, demand, land, ground, access and prices can remain provisional.
Engineering research	Counted vehicle and civil concepts, SolidCraft/FieldCraft studies, independent controls and source-bound candidate packets.	A calculated or converged model is not a qualified installation.
Business and maintenance	ERPNext/Frappe HR and FUXA/gateway adapters, draft records and a bounded assembly custody contract.	Native full-module transactions, live control and purchasing authority remain separate.

Use the capability register for specific entry points and maturity. The v0.6.1 known limitations define the published release boundary. The stale integrated engineering dossier and portfolio selector are held; this reader does not draw conclusions from them.

Development reference and research alternatives

The development reference uses six 17 m cars with passive wheel bogies on conventional rails carried by concrete. Each car has 44 fixed seats and two wheelchair places, with no planned standing and two doors on one side. The passive aluminium secondary is a comparison interface. Installed propulsion, powered length, acceleration, braking and operating headway are selected per candidate. The former maglev configuration and the 120-second service case are retained research controls, not the selected physical railway.

Research candidates may change traction, running gear, formation, station spacing or civil form. Each change must carry its mass, power, heat, protection, service, manufacture, construction and whole-life cost effects. Mandatory safety, accessibility and passenger protection stay as requirements. The architecture explains how candidate, evaluation and selection identities prevent results from being mixed.

One corridor through v0.7

The v0.7 roadmap uses one Samawah corridor revision as the integration case. Two source station markers are about 79.977 m apart against a 120 m platform assumption. The reviewed midpoint edit changes raw markers from 26 to 25; the published comparison already groups them into 25 operational stops, so its fleet stays at 81. The project envelope identifies which service, quantity, cost and dossier results change and which unaffected results are reused. These are planning calculations, not a surveyed alignment.

Current v0.7 evidence also includes a headless native GIS two-edit cycle for Samawah and Baghdad, bounded civil and force/braking controls, and a measured opt-in Rust delivery scheduler. The measured complete delivery-CLI median fell from 12.49 to 10.79 seconds across three paired Samawah trials with a prebuilt binary. That is a 13.64% reduction for this compiler path; it is not a whole-city or engineering-solver speedup. The agent delivery plan keeps further Rust work tied to measured bottlenecks and exact output parity.

The door module and the next physical interfaces

The candidate-bound one-opening handoff counts 74 part types and 694 pieces: 97 locally fabricated, 568 bought to standard and 29 bought to qualification. Its 597 purchased pieces span 35 types. The software custody contract checks receipt-backed, individually serialized purchased children with serial, batch, incoming QA and parent consumption. A sealed, reviewed synthetic one-child pack exercises that contract. The complete module has not passed a native ERPNext receipt, assembly, replacement and handback journey.

The opening layout still lacks controlled header/body, rail/carriage, belt/pulley, drive and sensor mating datums. Installed transforms, approved inspection limits, clearance and removal paths, complete installed mass and physical acceptance remain open. A 2 m display proxy does not establish fit for the 2.9 m guide stock. The handoff and evidence record identifies each unresolved interface; no missing quantity or limit is zero.

Civil work likewise needs a represented rail/support/load domain, bearings, foundations, erection stability and a crane chart at the actual lift radius. Propulsion and braking need installed force, current and repeated thermal-duty evidence. Independent physical tests and reviews remain necessary before any manufacturing, movement or operating release.

Use and reproduce the software

Start with the published installation instructions and their checksum-verified downloads. The Workbench shows supported journeys and visible holds. The repository exposes bounded research commands:

```
./osr design railway plan
./osr design railway run --max-bundles 2 --native-jobs 1
./osr design railway verify
./osr book --check
```

Native CAD, FEM and GIS jobs need compatible tools, declared resource budgets and exact input identities. Keep a host disk reserve and retain failed or interrupted run evidence. Current v0.7 work uses SolidCraft, FieldCraft and RustGIS in their demonstrated domains; optional independent solvers remain useful for verification. A tool substitution creates a new evaluation, not a new physical design by itself.

Source, review and history

The project README remains the public entry point. This PDF also includes that entry point, the capability register, release scope and limitations, and roadmap as source-attributed chapters. The v0.7 execution record remains linked and source-bound to this edition. The platform guide retains deeper v0.6 reference and research background under its stated evidence limits. The documentation index lists further maintained guides. The v0.7 execution ledger records task status, candidate boundaries and evidence. The current book PDF has its own source receipt; its summaries do not supersede the sealed engineering packets.

The published v0.6.1 release and historical v0.6.0, v0.5 and v0.4 records keep their original identities. Earlier maglev and source-bound research remain retrievable through the historical manifest. Their results apply only to their declared candidates and methods. Preserve useful failures, open assumptions and original hashes when revising the design.

Software is licensed under Apache 2.0, open physical designs under CERN-OHL-S v2, and documentation under CC-BY-SA 4.0, subject to path-specific notices in the license guide.

Project and products

README.md

OpenSourceRail develops affordable urban rail and open tools to plan, design, build, operate and renew it, keeping design knowledge, manufacture and repair capability in-country.

The v0.6 reference uses comfort cars on passive steel-wheel bogies, conventional rails and precast concrete. Station propulsion alternatives remain under comparison; no installed architecture or powered length is selected. This is a research and prototype design, without manufacturing or operating release.

Start here: current platform book · published v0.6.1 software · v0.7 execution and open findings · city designs and maps · engineering Workbench.

The v0.6 design-closure ledger retains seven engineering findings. The published v0.6.1 software scope and the v0.7 roadmap are separate from physical, manufacturing and operating acceptance. The older authenticated software-review profile remains historical and pending by design.

OpenSourceRail v0.6 proposed wheel-supported train at a compact elevated station

Concept. The retained v0.6 vehicle stock view and engineering models show partial geometry and its limits. The integrated current dossier has a source-binding hold and is excluded from supported v0.6.1 results.

What the project provides

Connected product	What it is for	Available today
Design and delivery tools	Turn a city brief into alternatives, engineering checks, costs and delivery records	Planning editor, calculated outlines, engineering Workbench and candidate exports
Reference railway and manufacturing system	Develop vehicles, stations, civil works, tooling, assembly and repair methods	Physical candidate definitions, CAD/BIM and numerical studies; physical qualification remains open
Open GoA4 control software	Develop protection, interlocking, station-led dispatch and train supervision	Rust research software, simulation and formal/method checks; no live railway or safety approval

Compare the whole installed system: vehicles, stations, civil works, factories, energy, maintenance and service availability. Comfort, safety and accessibility remain requirements.

Local capability includes precasting, fabrication, panels, interiors, wiring, assembly, inspection and repair. Local content requires origin evidence. Open drawings and records let owners change suppliers.

See the adoption path, capabilities and roadmap.

From a city brief to a working railway

Step	What to do	Where to start
1. Define the brief	Record demand, routes, access, climate, safety and local capability	City catalogue, planning workspace
2. Compare alternatives	Vary vehicles, propulsion, civil works, service and production under common requirements	Workbench, coordinated design workflow
3. Check service and cost	Calculate journeys, capacity, fleet, energy and low/base/high lifecycle costs	Demand-led outline, cost and delivery method
4. Plan delivery	Inspect CAD/IFC, BOM, process hours, procurement, construction and schedules	Candidate packages, pilot delivery compiler
5. Review, build and learn	Validate independently; configure production, commissioning, maintenance and measured feedback	ERPNext/Frappe HR, FUXA/gateway, platform guide

Revisions update one candidate through geometry, mass, checks, quantities and cost. Missing parts, hours and prices remain unknown or bounded.

The reconciled research pilot connects a proposed installed vehicle, axle loads, guideway, counted fleet, causal station energy recovery and fourteen-scope costs. A separately identified no-store alternative provides a whole-life comparison. Failed structural alternatives and unresolved engineering tests remain visible; the passive reference and city prices are unchanged.

Recovery cost comparison · Detailed bogie fit and assembly · Candidate-linked factory tasks. These research comparisons preserve unresolved physical interfaces and do not change the passive development reference.

Calculation, validation and release are separate. Calculations declare their assumptions. Validation needs physical evidence and independent review; manufacture and operation need their own acceptance.

Explore a city

The priced city and design outlines cover 266 mapped city networks and 1565 lines, plus 112 railway bundles, 54 bodies and 18 civil spans. The Workbench compares original, straighter and rounded square-ring proposals, joins close stop markers into operational complexes, and exposes fleet, manufacturing, construction and fourteen whole-life cost scopes. The local workflow edits and re-costs an outline, then exports a resource-leveled delivery plan and disabled business-system drafts. These are completed calculated outlines under explicit demand, site and rate assumptions; physical approval remains separate.

Start with Baghdad or Samawah for maps, service, fleet, energy, construction resources, costs and finance scenarios. Inspect routes, stations and calculations in the city railway atlas.

For Samawah, a current-source equal-propulsion route sensitivity preserves all 25 station markers while shortening the sampled line geometry by 5.64%. Its base CAPEX rises 9.27%, mainly because the simplified option carries a $97.93m unsurveyed access/crossing risk reserve. The report includes the assumptions, all fourteen cost scopes and a route-only map; this is a priced planning alternative, not a surveyed or selected alignment: study and results.

Baghdad now has a 14-line route sensitivity from its current published geometry: 13 radial lines shorten by 1.01% (5.016 km); all 157 operational stop coordinates remain fixed and the 26-stop ring is unchanged. Under a shared, assumed powered-axle research overlay, base CAPEX rises 10.44% and 30-year NPV 5.19%, mainly from an unverified corridor-risk allowance. The overlay has no supported force map, and the route is not surveyed. See the map, full cost comparison and reserve sensitivity.

In the Workbench, choose a priced city and calculate a scenario. Edit this outline and recalculate changes routes, stops and headways, updating map and figures together. Planning handoffs contain proposals. Site surveys, demand, ground, utilities, power, rescue and physical approval remain project work.

Worked costs: demand-led project, 30 km / 12 station benchmark and project-volume supply comparison. Compare these Class C estimates at equal service and compatible rates.

v0.6 reference vehicle

Item	Development basis
Formation and comfort	Six 17 m cars; 44 fixed seats + two wheelchair places/car; 276 places/train; no planned standing
Layout and body	Two passenger doors on one side; aluminium/GFRP architecture, sandwich floor/roof and lightweight interiors
Running gear and guideway	Twelve unpowered two-axle bogies, conventional rails and fastenings on concrete; box, twin-beam and U-beam comparisons
Propulsion and services	Passive aluminium secondary interface for LIM comparisons; separately sized service batteries, roof solar, HVAC, doors and emergency loads
Protection	Continuous onboard ATP, independent mechanical brakes and protected recovery/evacuation arrangements requiring qualification
Named service control	72 km/h launch cap, 120-second headway and 8,280 nominal places/hour/direction; this is a comparison case, not a required operating design

The reference selector excludes onboard traction motors, traction batteries and permanent magnets. Choose architecture, powered length, acceleration, braking and service per candidate against demand, physical constraints and whole-life cost. Neither 900 m nor 100 m is mandatory.

The current dossier records open mass, force, thermal, stopping, structural and installation evidence. The LIM interface supports comparison with mechanical launch and energy recovery and other launch alternatives.

Research may challenge the reference under the architecture campaign and transport-goal policy, with separate identities and complete system consequences. The former 120-person/car / 14,400 passengers/hour/direction comparison and 160-person/car crowd structural control remain named alternatives; seated service does not remove crowd, evacuation or accidental-load checks.

Models from CAD and analysis

These views come from recorded engineering geometry and native analysis results. Click a model for its design dossier. They show separate research candidates, with proposed stock and equipment envelopes identified; they do not select a combined installed railway.

Located vehicle cutaway	Bogie frame and drive cradle
One source-defined car with roof and near-side wall hidden to expose its frame, seating and equipment	Native SolidCraft view of the 34-solid bogie frame and drive-cradle stock subset
Stock and bounded equipment from the located research body; display-only cutaway.	Gross stock, with supplier housings and finished interfaces unresolved; fit HOLD.
Controlled-launch shuttle	Coupling subassembly
Native SolidCraft launch-shuttle assembly showing hollow members, guide wheels, bored clevis, pin and fixings	Native SolidCraft close-up of the bored clevis, pin and retainers, with surrounding stock ghosted
Actual 122-region CAD assembly; guards ghosted for inspection.	Pin, ears and retainers; contact, weld and fatigue checks remain open.
Proposed viaduct assembly	FieldCraft structural response
Source-defined two-track viaduct proposal with segments, rails, bearings, piers and footing envelopes	Actual FieldCraft displacement field on a 20 metre beam, with 100 times deformation and a millimetre legend
Located segment, rail and support proposal; joints and installed fit remain unverified.	Separate continuous elastic beam: 13.9476 mm midspan vertical displacement; shape exaggerated 100×.
Terminal seal joint	Body attachment bracket
Native SolidCraft terminal corner showing sealing lands, retainers, sleeves and panel fixings with surrounding sections ghosted	Native SolidCraft folded body attachment bracket with bored sleeves and fixings between ghosted frame sections
35 known parts represented by 43 material solids; seal and hardware details remain open.	Nine known parts represented by 11 material solids; joint strength and installed fit remain open.
Terminal frame fabrication	Welds and blind-bore detail
Native SolidCraft terminal side-frame member made from four plates and two endplates	Native SolidCraft close-up showing finite weld deposits and original endplate bores through ghosted plates
Six physical plates; 635 mm hollow member, 651 mm overall.	Exact local geometry checked in FreeCAD; welding access, strength and full-body installation remain open.
Revised direct-floor chair	Anchor hardware from below
Native SolidCraft chair proposal showing aluminium frame, sandwich shells, cushions, shared armrest and mounting feet	Native SolidCraft underside view exposing four anchor sleeves, insulating liners, backing pads and captive nut retainers

The revised local chair contains 275 material solids, independently checked with FreeCAD and 2,427 material/void probes. The views include nominal fastener blanks; the mating vehicle floor and support bridge are absent. Complete installed fit, loads and manufacturing approval remain open. Download the CAD, parts definition and reproducible checks.

Native SolidCraft chair frame with mounting feet, paired shell holes, crush sleeves and isolation pads

Retained earlier chair-frame detail: 66 proposed material solids, with independent FreeCAD geometry correspondence. Its mounting, access, strength and seat-height reconciliation remain open. Download CAD and inspect the checks.

The full-resolution FEM image uses the full native nodal field, including quadratic midside nodes, under one declared normal-base load and ideal bearing lines. It does not verify the segmented assembly. Render recipes, numerical data and reproduction commands include the vehicle image's render-resource hold and preserve the distinction between a calculated result and physical acceptance.

The railway in pictures

Passenger cabin	Modular vehicle assembly
Proposed seated passenger cabin with accessible door zone and wheelchair provision	Workshop assembly concept with body, roof, doors, glazing, seating and passive bogies
Precast guideway construction	Station LIM comparison
Crane placing a precast rail beam onto urban elevated piers	Passive wheel bogies and aluminium secondary between wayside LIM primaries

Station LIM comparison showing launch, freewheeling, receiving and independent protected stopping

The original renders and illustration brief explain the concept. The LIM picture and schematic select no installed system or powered length.

The 3D engineering gallery provides rotatable STL, coloured GLB, editable FreeCAD and STEP models, source identities and export checks.

The detailed launch-shuttle assembly adds bored clevises, a revised pin, hollow frame members, wheel/axle parts, guards and fasteners, with matching component FEM and fabrication estimates. The city price audit explains corrected energy accounts and separate civil-cost alternatives.

The lighting, inter-car services, coupler and rubber gangway audit traces the counted parts and provisional mass/cost rows. Their CAD fit, full-motion clearances, electrical flex duty and controlled drawings remain open.

Vehicle and workshop assembly	Rail and running-gear workshop
Exploded 17 m vehicle assembly with roof, seating, doors, passive bogies and proposed fixtures	Rail-finishing, wheelset assembly, bearing and passive-bogie inspection workstation
Capture braking energy	Controlled launch and acceleration
Cable arrestor drums, hydraulic vessels, isolation, cooling and controls	Guided launch shuttle, coupling and optional capstan booster drive
Viaduct erection	Precast manufacture
Crawler crane placing a source 20 m concrete box beam	Concrete box beam, reusable forms, void former and reinforcement jigs

The gallery distinguishes source geometry, equipment envelopes and proposed tooling. CAD agreement does not establish complete installed mass or physical acceptance.

Generated CAD and structural results

These views come from native SolidCraft span models and the matched FieldCraft / CalculiX comparison. The box image is the 54.720 t four-wall portfolio geometry; it is not the 55.704 t six-part beam with end diaphragms. The FEM plot compares two mesh levels for uncracked, perfectly bonded elastic models. It is a sensitivity comparison, not mesh independence or bridge qualification.

SolidCraft reference box	SolidCraft folded plate
Native SolidCraft four-wall reference box span	Native SolidCraft folded-plate span

Recorded FieldCraft and CalculiX coarse/fine displacement comparison, with model limits

The CAD, FEM and manufacturing dossier includes editable `.sc` and STEP files, source identities, scoped costs and the full analysis limits.

Delivery and operations tools

The vehicle parts catalogue breaks a separate six-car detail proposal into 605 named part types and 40 assemblies: bogies, frames, panels, roofs, chairs, doors, glazing, simple LED lighting, couplers and gangways. Browse materials, counted fixings, fabrication and assembly routes, inspection and scoped estimates through the Vehicle parts link in the Workbench. Detailed drawings, installed fit and release evidence remain open.

Candidate exports connect CAD/BIM, BOM, manufacturing, procurement, assembly, civil and controls records. Hours and costs remain provisional. The detailed guide explains process costing, factories, construction and whole-life scope.

ERPNext and Frappe HR provide configurable draft project, purchasing, stock, workforce, manufacturing, quality and maintenance workflows. FUXA and the gateway provide equipment observation, trends, alarms and review-linked lifecycle records through the integration service. Organisations, calendars, masters, equipment bindings and commissioning need configuration. Generated handoffs remain disabled drafts until reviewed.

The current native acceptance scenario has exercised QA, serialized manufacture, installation, gateway defect, outage recovery, repair and independent business review on disposable ERPNext/HR/FUXA services. It uses simulation assets; full workforce assignment, production handback enforcement and physical commissioning remain separate work.

See the capability register for current functions and the platform guide for detailed methods, integration boundaries and research sources.

Run locally

For the bounded software demonstration and release scope, start with the v0.6 quickstart, feature maturity table and known limitations. The quickstart does not install optional CAD/FEM/GIS services. The broader research and integration workflows below remain available for contributors.

```
./install.sh
./osr status
./osr readiness
./osr workbench
```

See setup for dependencies. Create and check a planning/factory handoff:

```
./osr plan compile edits.json --output build/my-plan
./osr plan verify build/my-plan
./osr twin factory-draft compile build/my-plan --output build/factory-plan
./osr twin factory-draft verify build/my-plan build/factory-plan
./osr twin export
```

`edits.json` is your proposal. Factory drafts require verified handoffs and remain unreleased. For bounded engineering and publication checks:

```
./osr design railway plan
```

```
./osr design railway run --max-bundles 2 --native-jobs 1
./osr design railway publish
./osr design railway verify
./osr build
./osr verify
./osr book --check
./osr test
```

Native workers need declared FEM/CAD tools and resources; runs retain failures and pending work. `./osr build` attempts to refresh the source-bound development packet, which is currently held by a source-binding mismatch. `./osr book` builds the current documentation reader independently of that held packet. The UI has no live train-control authority.

About the author

Hayder Aziz initiated OpenSourceRail. His research since the early 2000s connects product data management, manufacturing knowledge, open standards and collaborative software, bringing manufacturability and knowledge exchange into conceptual design.

His publications include New Technologies and Standards for Design Knowledge Management in Collaborative Conceptual Design (2003, with James Gao, Paul Maropoulos and Wai Ming Cheung), Open standard, open source and peer-to-peer tools and methods for collaborative product development (2005, Computers in Industry), and Engineering design knowledge management in de-centralised virtual enterprises (2008, Computer-Aided Design and Applications).

OpenSourceRail carries this research into a traceable transport design toolchain. Read the full biography and research context.

Contribute and learn more

Follow CONTRIBUTING, GOVERNANCE and project instructions. Contribute city data, supplier evidence, physical models, manufacturing/repair methods, controls or documentation. Edit sources and regenerate derived artifacts.

For deeper reading: platform guide · architecture · design engine · Rust controls · sourcing evidence · documentation index.

Software uses Apache 2.0, open physical designs use CERN-OHL-S v2, and documentation uses CC-BY-SA 4.0, subject to more-specific and upstream notices. See LICENSE.md and LICENSES/.

Current capabilities

docs/capabilities/README.md

OpenSourceRail connects three products: an open railway design and engineering system, a manufacturing/delivery and maintenance platform, and city planning and service tools. The aim is affordable whole-life transport supported by local design, manufacture and repair knowledge. Planning, engineering, business and equipment-observation records have distinct identities and review gates.

Reference and research configurations

The development baseline remains a six-car, wheel-supported passive reference: six 17 m cars, 44 fixed seats and two wheelchair places per car, with independently sized service batteries and brakes. It has no selected onboard traction installation. Speed, headway, propulsion architecture and powered length are candidate decisions; neither a 900 m LIM nor the historical 120-second schedule is mandatory.

The reconciled research pilot and no-store control are newer, separately identified powered-wheel research candidates. They join counted body parts, explicit external drives and reserves, axle loads, civil geometry, concurrent-train simulations, station storage alternatives and fourteen-scope cost ledgers. Several declared nominal timetables complete 16/16 model arrivals. This is conditional simulation evidence; it neither changes the passive reference nor selects an installed architecture for physical deployment.

Body mass subtotals, including a provisional installation reserve, remain separate from external motors, gearboxes, drive BOP and recovery fittings. Whole-train mass and cost must be read from the matching integrated evaluation. Proposed component positions, CAD volume comparisons and solver agreement do not establish measured installed mass, a collision-free assembly or qualified hardware. Native results apply only to their exact source, load, mesh and candidate; the no-store alternative does not inherit recovery-candidate FEM.

Available workflows

Capability	Current entry point and boundary
Coordinated candidate engineering	`./osr design railway`; preserves component/evaluation identities and open evidence gates
Reconciled research assembly and service	Recovery pilot and no-store control; conditional model-level arrivals, economics and fault cases
Native CAD and structural comparison	SolidCraft/FieldCraft pilot, viaduct alternatives and reconciled pilot; independent FreeCAD/CalculiX comparisons where recorded
Counted vehicle manufacturing definition	Vehicle parts catalogue and pilot installed-body records; materials, processes and stock quantities with detailed-drawing holds
Candidate-bound pilot manufacturing and installation proposals	Delivery compiler; local draft parts, source assembly routes, inspection holds, external ownership and existing cost-scope references
Selected reference mass account	Mass reconciliation and roof-finish distribution estimate; distinct from powered-wheel research mass
Reference CAD/IFC, BOM, procurement, assembly and control records	Current export manifest and delivery package index
City geography, maps and route/service proposals	Planning guide, workspace; `./osr plan compile` / `verify`; route-specific engineering remains separate
Selected-candidate factory work package	Current Workbench; `./osr twin factory-draft compile` / `verify`; planning draft from the selected Class C ledger
Calculated manufacturing and lifecycle scenarios	Selected 6 km system estimate, vehicle estimate, cost and delivery method, and separate pilot ledgers
Business and workforce workflows	ERPNext/Frappe HR; retained integration contracts and native business-platform acceptance scenario, with reviewed conversion required for the new pilot
Equipment observation and maintenance review	FUXA/gateway; observations and reviewed maintenance handoffs, without safety movement authority
Open control software	Rust workspace; software tests remain distinct from physical qualification

The pilot delivery compiler preserves part definitions, engineering revisions, occurrence patterns and future serial/lot allocation separately. Its ready-for-planning-review tasks support material/route review and preparation of supplier information requests. Production, installation, procurement release and ERP conversion remain held for actual controlled drawings, reviewed native Item mappings, inspection references and engineering evidence. No accepted serial installation or approved work order is generated. Its process-hour ranges are declared capacity hypotheses, not measured cycle times or a second labour charge on existing direct prices.

The retained factory-draft command creates a six-car work package from a verified planning handoff and the selected candidate's Class C ledger, under two 8-hour production shifts. Its schedule represents one serialized lane and does not infer crews, parallel capacity or dates. It is not an ERPNext import, purchase order or manufacturing release. The planning handoff itself contains route, stop, service and OD proposals; vehicle, propulsion, fleet, civil, financial and performance evaluations remain separate candidate-specific records.

The selected reference ledger retains a 10.090 t/car reference or unlocated residual. Its separate roof-finish estimate distributes already-accounted 47.809 kg/car over 45.05 m² of source surface; it adds no mass or CAD occurrence. Those reference-scope figures must not be combined with research-pilot totals.

Preserved platform and open evidence

The v0.4 operating/manufacturing release and v0.5.1 maglev research remain distinct historical foundations. ERPNext/Frappe HR, FUXA/gateway, revision-controlled assets, city planning, workforce and maintenance are retained platform capabilities, not replaced by a new mechanical candidate. The source-pinned historical manifest records preserved engineering evidence. Historical results do not qualify a new wheel-supported design.

Calculated outlines can be complete under stated assumptions. Installed interfaces, actual force/duty and thermal behaviour, independent braking, structural joints/fatigue, site-specific works, controlled production drawings, physical prototypes and independent acceptance remain open gates. The new pilot's drafts do not yet constitute a seamless approved ERPNext project or an operational commissioning workflow. Software functionality grants no manufacturing release, physical acceptance or movement authority.

Published v0.6.1 software

docs/releases/v0.6.1/README.md

Published on 12 November 2026: v0.6.1 release. The immutable tag identifies commit `5cfe89488a64549b37d4a7fd3ea9e3d3a8923aea` and tree `58ba4411d502a166b3a14000bc1e09b733b65cbd`.

Start with the published INSTALL-v0.6.1.md. Download the release assets and verify them against SHA256SUMS. The quickstart follows that tagged installation route; native CAD/FEM/GIS and business-system services are optional for the Samawah demo.

Document	Purpose
Quickstart	Install the published tag and run the Samawah journey
Demonstrations	Samawah, Baghdad and candidate research walkthroughs
Feature maturity	Available, preview, held and planned functions
Release notes	Published scope and download links
Validation	Exact-tag CI evidence and clean-install handover procedure
Known limitations	Current dossier, native GIS and engineering boundaries

The software release includes planning, engineering research and delivery tools. Its Samawah journey calculates and exports a fresh case, reuses the identical request, and gives changed inputs a distinct evaluation. Baghdad and candidate examples use retained source-pinned packets. The current integrated dossier and portfolio selector remain visibly held and excluded from supported results; the native GIS round trip is v0.7 work.

The older design-closure campaign is superseded where it controlled software publication. Its seven open/partial engineering findings remain in the engineering ledger and feed the v0.7 roadmap. The retained authenticated-review profile keeps its truthful pending records; it does not describe whether v0.6.1 is published, and no passing review was fabricated.

The earlier annotated `v0.6.0` development tag remains at `61080cafc6998e587772810707dc552524df5306`; its historical preparation documents are retained. Neither release tag is moved by this handover. Physical, manufacturing, movement and operating acceptance remain separate from software publication.

The published installation verification records the completed clean installation and two Samawah journeys against the tag.

Published limitations

docs/releases/v0.6.1/known-limitations.md

- The passive-wheel development reference remains a design reference, not a physically accepted installation. The installed propulsion architecture is not selected.
- Vehicle mass, equipment fit, fasteners, joints, cooling, wiring, clearances, fatigue and removal access have unresolved bounds or holds.
- Propulsion force, current, heat, braking and rescue evidence is candidate- and model-domain-specific. Numerical convergence does not establish feasibility.
- City demand, route geometry, gradients, ground, utilities, land and access may be generic or unsurveyed. Cost ranges are not vendor quotations or approved finance.
- The 266-city catalogue is a planning portfolio. It is not 266 independently surveyed and engineered railway designs.
- The integrated current engineering dossier currently has a source-binding hold. The Workbench shows the selected reference and city geography, but keeps global engineering cost and portfolio views held and does not treat the stale dossier as current evidence. The dossier is excluded from supported v0.6 results and release evidence until its exact provenance is resolved. At this source commit, the portfolio hold identifies `.github/workflows/development-baseline.yml` as changed from the dossier's recorded source hash. The Workbench therefore hides its candidate-portfolio selector; use the source-pinned archive described in the demonstrations document for read-only research comparison. The Samawah software journey binds and verifies its own city inputs and calculation.
- The city atlas currently refuses native RustGIS ZIP export because the published GIS and outline epochs differ. The source-bound Baghdad result and its read-only atlas view remain available; a native GIS export/import round trip is not included in v0.6.
- CAD/IFC stock agreement and structural screens do not establish complete installed mass, manufacturability, fatigue life or construction approval.
- ERPNext, Frappe HR, FUXA and gateway workflows are configurable software examples. Real transactions and live railway commands are not authorized by generated drafts.
- Full native CAD/FEM/GIS workflows are optional and limited to declared solver versions, resource requests, inputs and model domains.
- Large historical studies remain in the repository history. Sparse checkout and optional local cleanup reduce active workspace use, but no repository history rewrite or complete artifact migration has been performed.
- Physical prototype validation, independent safety assessment and any manufacturing, movement or operating approval remain separate future work.

v0.7 roadmap

docs/ROADMAP.md

v0.6.1 — Open Engineering Platform is the published v0.6-series software release at `5cfe89488a64549b37d4a7fd3ea9e3d3a8923aea`. Use its installation route, downloads and known limitations. The existing `v0.6.0` and `v0.6.1` tags stay at their recorded commits.

OpenSourceRail continues to connect railway design and engineering, manufacturing/delivery and maintenance, and city planning and service. Local manufacture, repairability, open knowledge, ERPNext/Frappe HR and FUXA/gateway integration remain part of the platform's purpose.

Finish the v0.6 handover

The handover is complete. A fresh public clone of the published tag followed its installation asset, verified SHA256SUMS and completed the documented Samawah journey twice. Installation results record commands, identities, timings and the environment limits. The retained readiness-browser harness now uses an external temporary output directory; its repaired CI job passed on main `0472a4f3cadb23847e071e693294a418bf54c3b0`.

The older design-closure campaign is superseded only where it controlled software publication. Its seven open/partial engineering findings remain visible in the engineering ledger. The retained authenticated-review profile keeps its truthful pending records; it is not a reason to repeat publication checks or withhold the already published release.

First v0.7 milestone: the Samawah station merge

Open the published Samawah example, apply the retained station-merge intent, and produce a side-by-side report of station count, service assumptions, fleet, civil quantities, fourteen-scope costs and unresolved inputs. Each difference must trace to the edit and its dependent calculation. Reuse unaffected vehicle and manufacturing outputs rather than rerunning them.

The retained engineering case found two station markers 79.977 m apart against a 120 m platform assumption. Its midpoint merge packet provides the before/after edit and provenance. Carry both cases through the new workflow. That packet belongs to its declared candidate/source epoch: applying the edit to the published example must produce and identify the matching new revision, not silently transfer old compact-drive results to another vehicle. Access, catchment, platform fit, route feasibility and unpriced work remain explicit inputs or findings.

The first Workbench comparison is implemented and browser-verified. The published case already groups the close pair operationally: raw markers change from 26 to 25, operational stops remain 25 and fleet remains 81. Both sides retain the same vehicle candidate and unchanged civil quantity identity; the report exposes actual service, access and fourteen-scope cost changes. The remaining workstreams below remain open.

The agent delivery and tool migration plan assigns bounded work to agents and subagents, with Rust migration benchmarks, CAD/FEA/GIS promotion criteria, legacy retirement gates and resource budgets. The execution record now contains the verified two-worker improvement, opt-in Rust scheduler, fresh CAD/FEA controls, native Samawah/Baghdad cycles and clean-install CI evidence. Its task ledger keeps the remaining installed-candidate, detailed module and lifecycle work open.

v0.7 delivery workstreams

Workstream	Required integrated result
Project continuity	One project entry point binds corridor revision, vehicle candidate, civil option, quantities, cost and dossier. A station edit invalidates its dependants; the Workbench shows changed results and verified reuse of unaffected work.
Installed vehicle	Select one six-car candidate for detailed integration. Resolve recurring geometry definitions, door motion, equipment access/removal and mass bounds. Keep the passive reference and propulsion alternatives separate; supplier and measurement dependencies remain explicit.
Civil works and station	Carry one span, support and station interface through bearing assumptions, moving loads, construction sequence and lifting checks. Establish crane capacity at the actual installation radius. Retain SolidCraft/FieldCraft and independent FreeCAD/CalculiX evidence.
GIS and economics	Complete a fresh native GIS edit–export–import–reprice cycle for the selected Samawah revision. Trace station count, fleet, guideway quantities and all fourteen cost scopes; identify transferred rates and unpriced quantities. Check the implementation on one Baghdad corridor afterward.
Manufacture and lifecycle	Use the door operator as the representative module. Bind drawings, tolerances, BOM, assembly/tooling and QA criteria to its geometry and revision; exercise removal/replacement, wiring and motion clearance. Run that actual candidate-linked module through the existing ERPNext/FUXA lifecycle rather than inheriting the synthetic one-unit kit result.
Workbench and release	Guide one corridor edit through service, cost and revised dossier, with links to calculations, drawings and open findings. Open the same module for manufacture planning, QA and maintenance. Preserve before/after outputs and replay the full journey from a clean install.

Delivery order and acceptance evidence

1. Handover and baseline — complete: the published v0.6.1 installation, Samawah before/after comparison, candidate boundaries and bounded performance baseline have recorded source identities and timings.
2. Project continuity — software demonstrated: the station edit flows through service, fleet and fourteen cost scopes with dependency checks and visible missing installed stages. Complete candidate integration remains open.
3. Engineering interfaces — active: establish controlled door mating datums, force/brake/thermal inputs and represented civil support/loading before new native CAD or FEA. The Samawah and Baghdad native GIS two-edit return cycles are recorded; rerun only for changed inputs.
4. Lifecycle and generalisation — active: take the counted door module through actual native ERPNext receipt, QA, assembly, replacement and handback, then compile a same-candidate dossier and replay one Baghdad corridor.
5. Release acceptance: resolve defects exposed by both journeys, run the clean-install v0.7 acceptance case, and publish supported scope and remaining engineering limitations.

The original week ranges were planning estimates and are superseded by the execution ledger. Supplier inputs, physical measurements and unresolved structural assumptions have separate milestones; software can show exactly which conclusions depend on them.

Consolidate existing computation

Use selected-city source binding, verified caches and incremental publication checks already in the repository. Measure wall time by stage at the corridor entry point. If a station edit launches an unchanged vehicle analysis, repair the dependency first. Benchmark an expensive required kernel before extending Rust or vectorisation. Avoid broad rewrites, all-city regeneration and duplicate calculation pipelines for this milestone. Preserve failed, interrupted and out-of-domain evidence and maintain the project disk reserve during execution.

Retained engineering foundation

The reconciled recovery pilot and no-store control connect an installed-body hypothesis, separately owned drive/BOP additions, mass and axle loads, civil design, concurrent trains, station energy allocation and fourteen-scope costs. Several nominal model timetables complete 16/16 arrivals. Source-linked CAD and independent structural comparisons are retained within their declared applicability. These achievements close important software and calculated-outline links; equipment characteristics remain provisional.

The selected development reference remains passive and wheel-supported. The newer powered-wheel pilots are research alternatives, with separate physical and evaluation identities. No installed propulsion architecture is physically selected. Body-plus-reserve subtotals exclude external motors, gearboxes, drive BOP and recovery interfaces; integrated mass and cost must come from the same candidate. A child revision must propagate through all affected evaluations rather than inherit its parent's results.

Coordinated work continues through `./osr design railway`, the cost and delivery method, pilot delivery compiler, ERPNext/FUXA lifecycle and capability register. Retain the v0.4 platform, v0.5.1 research and source-bound archives independently reproducible.

Cost, resource and production assumptions

Keep inexpensive screening ahead of detailed CAD/FEM. Reuse numerical work only for the exact source, tool, physical candidate, load, mesh and resource request. Run independent cases within measured memory budgets, preserve failed runs and benchmark useful candidates per compute-hour. Python remains useful for orchestration and specialist integration.

Current outline scenarios use two 8-hour production shifts and two 8-hour civil shifts. Operating cost uses two shifts for 16/18-hour service and three for the 24-hour case. Utilisation, shift premiums, relief staffing and wages remain explicit hypotheses. Equivalent resource hours do not establish an actual crew, production calendar, construction critical path or approved roster.

Every price retains quantity, source or assumption, scope, uncertainty and exclusions. Factory, delivered, installed and lifecycle totals remain distinct. Direct part/process allowances, external reserves and fourteen-scope accounts must reconcile without counting the delivery decomposition or capacity proxies again. China sourcing, bulk procurement and local manufacture can be compared using explicit production/supply evidence; an automatic discount is not evidence.

Open physical engineering gates

Area	Required next evidence
Vehicle	Installed mounting and movement geometry, remaining equipment, qualified joints/running gear, measured or validated mass/CG/inertia
Propulsion and recovery	Attainable force-speed-duty envelopes, capture/reset/release losses, thermal limits, independent stopping and recovery
Electrical and HVAC	Hot-weather charge/discharge/converter limits, cooling, emergency endurance, fire/isolation and maintenance access
Protected service	Model bounds plus validated braking/WSP, degraded operation, terminal protection, separation, rescue and actual controls
Civil and construction	Prestress/joints/cracking/fatigue, bearing/contact and dynamic behaviour, site ground, temporary works and verified lift/transport configurations
Production and business	Controlled drawings, tolerances, materials/process qualification, QA hold points, reviewed ERP conversion, actual capacity and traceable installation
Economics and site	Equal-service factory-to-installed/lifecycle ranges, renewals/recovery/disposal, local demand, utility/land/ground and supplier evidence
Prototype and release	Authenticated measurements, model correlation, independent review and separate manufacturing/operating approvals

Software completion, numerical convergence and a favourable estimate do not substitute for physical evidence or independent acceptance. Draft tasks and calculated outlines remain useful completed design-stage results.